

Common Mistakes Intermediate

A checklist worth reviewing before every regression project you ship.

USING R^2 ALONE

Correct thinking: always pair R^2 with an error metric in real units (MAE/RMSE) and a residual plot.

EVALUATING ONLY ON TRAINING DATA

Correct thinking: always hold out a genuine test set before trusting any performance number.

ASSUMING CORRELATION MEANS CAUSATION

Correct thinking: a coefficient describes association within the model, not a proven causal mechanism.

IGNORING RESIDUAL PATTERNS

Correct thinking: a good aggregate score can still hide a systematically wrong sub-region of the data.

IGNORING OUTLIERS

Correct thinking: check whether outliers are data errors, genuinely rare events, or influential points pulling the whole line.

LEAKING TEST DATA

Correct thinking: fit any scaler, encoder, or imputer only on the training set, then apply it unchanged to the test set.

INTERPRETING COEFFICIENTS INCORRECTLY

Correct thinking: always read a coefficient as "holding other features constant" — never in isolation.

ASSUMING LINEAR RELATIONSHIPS EVERYWHERE

Correct thinking: check the scatter plot and residuals — reach for polynomial terms or another model family when the shape is clearly curved.

BLINDLY REMOVING FEATURES BECAUSE OF CORRELATION

Correct thinking: correlation with the target is good; correlation *between features* is the concern — don't confuse the two.

CONFUSING MAE AND RMSE

Correct thinking: RMSE penalizes large errors more heavily; pick based on whether large mistakes are especially costly.

THINKING LINEAR REGRESSION MEANS ONLY ONE FEATURE

Correct thinking: multiple regression (Ch. 10), polynomial terms (Ch. 22), and encoded categories (Ch. 21) are all still "linear regression."

ASSUMING HIGHER R^2 AUTOMATICALLY MEANS A BETTER MODEL

Correct thinking: R^2 mechanically rises with added features — check adjusted R^2 and test-set performance instead.